ArchMind AI

Platform Database Architecture & Entity Relationship Diagram (ERD)

Comprehensive Technical Specification • MongoDB Atlas • Mongoose ODM • Clerk Auth

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COLLECTIONS	EMBEDDED SUBDOCS	TOTAL RELATIONSHIPS	DATABASE INDEXES
7 Models	6 Entities	15 Mappings	35 Indexes

1. Executive Database Overview

ArchMind AI utilizes a multi-model document database architecture powered by MongoDB Atlas and Mongoose ODM. The database is structured to balance multi-tenant workload isolation, ultra-fast read retrievals for interactive codebase exploration, and defense against MongoDB's 16MB BSON document limit through explicit file-decoupling.

Multi-tenant data partitioning is enforced via Clerk Authentication (User.clerkId). High-cardinality entities such as SourceFile documents and granular SecurityFinding records are persisted in dedicated, indexed collections with compound uniqueness constraints, while hierarchical structures (file trees, validation checks, chat citations) are efficiently embedded.

2. ERD Notation & Cardinality Legend

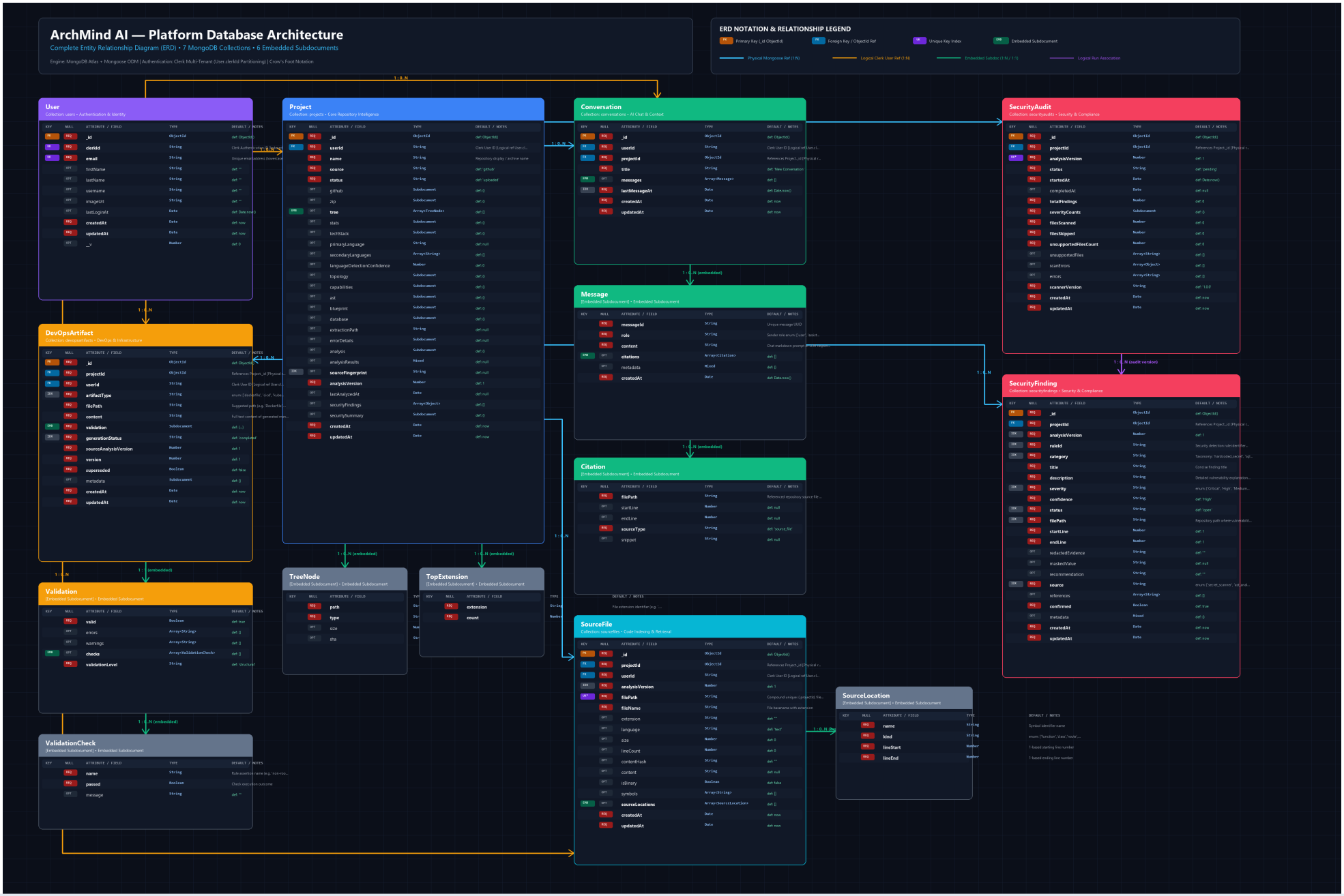
SYMBOL	MEANING	CARDINALITY	ARCHITECTURAL ROLE
PK	Primary Key	**1 : 1**	MongoDB _id auto-assigned unique BSON ObjectId.
FK (Phys)	Foreign Key (Physical)	**1 : 0..N**	Mongoose ObjectId ref referencing parent collection.
FK (Log)	Foreign Key (Logical)	**1 : 0..N**	Application-level tenant key referencing User.clerkId.
UK	Unique Key	**1 : 1 / Compound**	Enforced database index preventing duplicate entities.
EMB	Embedded Subdocument	**1 : N (Embedded)**	Denormalized array or sub-schema inside parent document.
---o{	Crow's Foot Zero/Many	**0..N**	Optional child participation (e.g. Project has 0..N audits).
---||	Crow's Foot Exactly One	**1 : 1**	Mandatory parent participation (e.g. Conversation has 1 project).

3. Core Functional Domains

Authentication & Identity	Clerk identity syncing, user profiles, login timestamps.
Core Repository Intelligence	Ingestion state, file tree, tech stack, AST blueprint, topology.
AI Chat & Grounded Context	Threaded repository chat, tokens, validated line citations.
Code Indexing & Retrieval	Decoupled file storage avoiding 16MB limit, symbol spans.
Security & Compliance	Secret scan runs, AST vulnerability rules, redacted evidence.
DevOps & Infrastructure	Dockerfiles, CI/CD pipelines, Kubernetes, syntax validation.

ArchMind AI — Complete Platform Entity Relationship Diagram (ERD)

Visual Overview of 7 MongoDB Collections, 6 Embedded Entities, and Crow's Foot Relationships



User Collection

Collection: users • Domain: Authentication & Identity

Mongoose Options: timestamps: true, versionKey: true

Description:

Stores user account profiles, Clerk authentication mappings, identity details, and login audit timestamps. Acts as the root tenant identity.

Database Indexes & Constraints:

- _id (PK)
- clerkId (unique)
- email (unique)

Attribute Definitions:

KEY	NULL	FIELD NAME	TYPE	DESCRIPTION & DEFAULTS
_id	ObjectId	PK	REQ	Auto-generated unique BSON identifier [def: ObjectId()]
clerkId	String	UK	REQ	Clerk Authentication user ID (e.g. 'user_2...') [def: Unique index]
email	String	UK	REQ	Unique lowercase email address (trimmed) [def: Unique index]
firstName	String	---	OPT	Given name synchronized from Clerk profile [def: ""]
lastName	String	---	OPT	Family name synchronized from Clerk profile [def: ""]
username	String	---	OPT	User-selected or Clerk handle [def: ""]
imageUrl	String	---	OPT	Avatar profile picture URL [def: ""]
lastLoginAt	Date	---	OPT	Timestamp of most recent authentication / sync [def: Date.now()]
createdAt	Date	---	REQ	Document creation audit timestamp [def: Mongoose timestamps]
updatedAt	Date	---	REQ	Last document modification audit timestamp [def: Mongoose timestamps]
__v	Number	---	OPT	Internal Mongoose document version key [def: 0]

Project Collection

Collection: projects • Domain: Core Repository Intelligence

Mongoose Options: timestamps: true, versionKey: false

Description:

Central repository metadata hub storing file trees, tech stack, topology graphs, Tree-sitter AST, prompt blueprints, and multi-module analysis states.

Database Indexes & Constraints:

- _id (PK)
- { userId: 1, 'github.fullName': 1 } (unique, partialFilter: source='github')
- { userId: 1, 'zip.originalName': 1 } (unique, partialFilter: source='zip')
- userId (asc)
- status (asc)
- sourceFingerprint (asc)

Attribute Definitions:

KEY	NULL	FIELD NAME	TYPE	DESCRIPTION & DEFAULTS
_id	ObjectId	PK	REQ	Auto-generated unique BSON identifier [def: ObjectId()]
userId	String	FK	REQ	Clerk User ID [Logical Reference to User.clerkId] [def: Indexed]
name	String	---	REQ	Repository display name or archive title [def: Trimmed]
source	String	---	REQ	Ingestion origin enum: ['github', 'zip'] [def: 'github']
status	String	IDX	REQ	Pipeline lifecycle: ['uploaded','processing','completed','failed'] [def: 'uploaded']
github	Subdocument	---	OPT	GitHub metadata: { githubId, owner, repo, fullName, defaultBranch, ... } [def: {}]
zip	Subdocument	---	OPT	ZIP upload metadata: { originalName, sizeBytes } [def: {}]
tree	Array<TreeNode>	EMB	OPT	Embedded sanitized repository file hierarchy [1:N TreeNode] [def: []]
stats	Subdocument	---	OPT	Summary metrics: { totalFiles, totalDirs, totalBytes, topExtensions } [def: {}]
techStack	Subdocument	---	OPT	Detected stack: { languages: [], frameworks: [], databases: [], devops: [] } [def: {}]
primaryLanguage	String	---	OPT	Dominant language identified by RIE [def: null]
secondaryLanguages	Array<String>	---	OPT	Secondary languages identified in repository [def: []]
languageDetectionConfidence	Number	---	OPT	Statistical confidence score of language identification [def: 0]
topology	Subdocument	---	OPT	React Flow architecture graph: { nodes: [], edges: [], generatedAt } [def: {}]
capabilities	Subdocument	---	OPT	Relevance matrix: { hasDatabase, hasApis, hasDevops, details } [def: {}]
ast	Subdocument	---	OPT	Tree-sitter structural AST: { summary, files, routes, models, classes, ... } [def: {}]
blueprint	Subdocument	---	OPT	Ground-truth LLM prompt blueprint: { version, routes, models, tokenEst } [def: {}]
database	Subdocument	---	OPT	Static schema extraction: { type, entities: [], relationships: [] } [def: {}]
extractionPath	String	---	OPT	Ephemeral extraction path in server/storage/extracted [def: null]
errorDetails	Subdocument	---	OPT	Diagnostic on pipeline failure: { message, code, at } [def: null]
analysis	Subdocument	---	OPT	Per-module state: { architecture, erd, api, security, devops } [def: {}]
analysisResults	Mixed	---	OPT	Cached downstream AI analysis payloads [def: null]
sourceFingerprint	String	IDX	OPT	SHA-256 fingerprint of repository content for caching [def: Indexed]
analysisVersion	Number	---	REQ	Sequential analysis version counter [def: 1]

KEY	NULL	FIELD NAME	TYPE	DESCRIPTION & DEFAULTS
lastAnalyzedAt	Date	---	OPT	Timestamp of most recent analysis run [def: null]
securityFindings	Array<Object>	---	OPT	Denormalized projection of security findings for fast UI render [def: []]
securitySummary	Document	---	OPT	Aggregate severity counts: { Critical, High, Medium, Low } [def: {}]
createdAt	Date	---	REQ	Document creation audit timestamp [def: Mongoose timestamps]
updatedAt	Date	---	REQ	Last document modification audit timestamp [def: Mongoose timestamps]

Conversation Collection

Collection: conversations • Domain: AI Chat & Grounded Context

Mongoose Options: timestamps: true, versionKey: false

Description:

Repository AI chat conversations preserving prompt histories, token metadata, and verified line-level source code citations.

Database Indexes & Constraints:

- _id (PK)
- { userId: 1, projectId: 1, updatedAt: -1 } (compound search index)
- userId (asc)
- projectId (asc)
- lastMessageAt (asc)

Attribute Definitions:

KEY	NULL	FIELD NAME	TYPE	DESCRIPTION & DEFAULTS
_id	ObjectId	PK	REQ	Auto-generated unique BSON identifier [def: ObjectId()]
userId	String	FK	REQ	Clerk User ID [Logical Reference to User.clerkId] [def: Indexed]
projectId	ObjectId	FK	REQ	Target repository [Physical Reference to Project._id] [def: Indexed]
title	String	---	REQ	Conversation title (maxlength: 120 chars) [def: 'New Conversation']
messages	Array<Message>	EMB	OPT	Embedded chronological messages [1:N Message] [def: []]
lastMessageAt	Date	IDX	REQ	Timestamp of most recent message [def: Date.now()]
createdAt	Date	---	REQ	Document creation audit timestamp [def: Mongoose timestamps]
updatedAt	Date	---	REQ	Last document modification audit timestamp [def: Mongoose timestamps]

SourceFile Collection

Collection: sourcefiles • Domain: Code Indexing & Retrieval

Mongoose Options: timestamps: true, versionKey: false

Description:

Indexed source file storage separate from the Project document to prevent MongoDB's 16MB document limit and enable citations.

Database Indexes & Constraints:

- _id (PK)
- { projectId: 1, filePath: 1, analysisVersion: 1 } (unique)
- projectId (asc)
- userId (asc)
- analysisVersion (asc)

Attribute Definitions:

KEY	NULL	FIELD NAME	TYPE	DESCRIPTION & DEFAULTS
_id	ObjectId	PK	REQ	Auto-generated unique BSON identifier [def: ObjectId()]
projectId	ObjectId	FK	REQ	Target repository [Physical Reference to Project._id] [def: Indexed]
userId	String	FK	REQ	Clerk User ID [Logical Reference to User.clerkId] [def: Indexed]
analysisVersion	Number	IDX	REQ	Sequential analysis version counter [def: 1]
filePath	String	UK*	REQ	Relative path within repository (compound unique) [def: Trimmed]
fileName	String	---	REQ	Basename of file including extension [def: Required]
extension	String	---	OPT	File extension (e.g. '.js', '.py') [def: ""]
language	String	---	OPT	Programming language identifier [def: 'text']
size	Number	---	OPT	Raw file size in bytes [def: 0]
lineCount	Number	---	OPT	Total lines of source code [def: 0]
contentHash	String	---	OPT	SHA-256 hash of file content [def: ""]
content	String	---	OPT	Sanitized & secret-redacted source text [def: null if binary/large]
isBinary	Boolean	---	OPT	Flag indicating binary/non-text file [def: false]
symbols	Array<String>	---	OPT	Extracted code symbols (functions, classes) [def: []]
sourceLocations	Array<SourceLocation>	EMB	OPT	AST line spans [1:N SourceLocation] [def: []]
createdAt	Date	---	REQ	Document creation audit timestamp [def: Mongoose timestamps]
updatedAt	Date	---	REQ	Last document modification audit timestamp [def: Mongoose timestamps]

SecurityAudit Collection

Collection: securityaudits • Domain: Security & Compliance

Mongoose Options: timestamps: true, versionKey: false

Description:

Persistent security audit run metadata tracking scan coverage, severity tallies, scanner versions, and unsupported files.

Database Indexes & Constraints:

- _id (PK)
- { projectId: 1, analysisVersion: 1 } (unique)
- projectId (asc)
- analysisVersion (asc)
- status (asc)

Attribute Definitions:

KEY	NULL	FIELD NAME	TYPE	DESCRIPTION & DEFAULTS
_id	ObjectId	PK	REQ	Auto-generated unique BSON identifier [def: ObjectId()]
projectId	ObjectId	FK	REQ	Target repository [Physical Reference to Project._id] [def: Indexed]
analysisVersion	Number	UK*	REQ	Analysis run version (compound unique with projectId) [def: 1]
status	String	IDX	REQ	Lifecycle: ['pending','processing','completed','failed'] [def: 'pending']
startedAt	Date	---	REQ	Timestamp when security audit scan began [def: Date.now()]
completedAt	Date	---	OPT	Timestamp when security audit scan ended [def: null]
totalFindings	Number	---	REQ	Aggregate count of detected security issues [def: 0]
severityCounts	Subdocument	---	REQ	Severity tallies: { Critical, High, Medium, Low } [def: {}]
filesScanned	Number	---	REQ	Total repository files scanned for security issues [def: 0]
filesSkipped	Number	---	REQ	Total repository files ignored by noise filter [def: 0]
unsupportedFilesCount	Number	---	REQ	Count of unsupported/binary file formats [def: 0]
unsupportedFiles	Array<String>	---	OPT	List of skipped non-analyzable files [def: []]
scanErrors	Array<Object>	---	OPT	Diagnostic error records from scanners [def: []]
errors	Array<String>	---	OPT	High-level execution error messages [def: []]
scannerVersion	String	---	REQ	ArchMind security engine release version [def: '1.0.0']
createdAt	Date	---	REQ	Document creation audit timestamp [def: Mongoose timestamps]
updatedAt	Date	---	REQ	Last document modification audit timestamp [def: Mongoose timestamps]

SecurityFinding Collection

Collection: securityfindings • Domain: Security & Compliance

Mongoose Options: timestamps: true, versionKey: false

Description:

Deterministic secret detections, Tree-sitter AST security vulnerabilities, and AI code review findings with redacted evidence.

Database Indexes & Constraints:

- _id (PK)
- { projectId: 1, analysisVersion: 1, status: 1 } (compound filter)
- { projectId: 1, analysisVersion: 1, severity: 1 } (compound triage)
- { projectId: 1, filePath: 1 } (file finding lookup)
- projectId (asc)
- analysisVersion (asc)
- ruleId (asc)
- category (asc)
- severity (asc)
- status (asc)
- source (asc)

Attribute Definitions:

KEY	NULL	FIELD NAME	TYPE	DESCRIPTION & DEFAULTS
_id	ObjectId	PK	REQ	Auto-generated unique BSON identifier [def: ObjectId()]
projectId	ObjectId	FK	REQ	Target repository [Physical Reference to Project._id] [def: Indexed]
analysisVersion	Number	IDX	REQ	Associated SecurityAudit analysisVersion [def: 1]
ruleId	String	IDX	REQ	Rule code identifier (e.g. 'SEC-SECRET-01') [def: Indexed]
category	String	IDX	REQ	Category: 'hardcoded_secret', 'sql_injection', etc. [def: Indexed]
title	String	---	REQ	Concise finding title [def: Required]
description	String	---	REQ	Comprehensive vulnerability explanation & threat analysis [def: Required]
severity	String	IDX	REQ	Severity enum: ['Critical', 'High', 'Medium', 'Low'] [def: Indexed]
confidence	String	---	REQ	Confidence enum: ['High', 'Medium', 'Low'] [def: 'High']
status	String	IDX	REQ	Triage status enum: ['open', 'resolved', 'ignored'] [def: 'open']
filePath	String	IDX	REQ	Vulnerable file path within repository [def: Trimmed]
startLine	Number	---	REQ	1-based starting line of vulnerable code [def: 1]
endLine	Number	---	REQ	1-based ending line of vulnerable code [def: 1]
redactedEvidence	String	---	OPT	Sanitized source code snippet (secrets redacted) [def: ""]
maskedValue	String	---	OPT	Partially masked token (e.g. 'ghp_****') [def: null]
recommendation	String	---	OPT	Concrete engineering remediation advice [def: ""]
source	String	IDX	REQ	Detection engine: ['secret_scanner','ast_analyzer','ai_reviewer'] [def: Indexed]
references	Array<String>	---	OPT	CWE, OWASP, and NIST vulnerability references [def: []]
confirmed	Boolean	---	REQ	Evidence confirmation verification flag [def: true]
metadata	Mixed	---	OPT	AST coordinates, entropy metrics, scanner output [def: {}]

KEY	NULL	FIELD NAME	TYPE	DESCRIPTION & DEFAULTS
createdAt	Date	---	REQ	Document creation audit timestamp [def: Mongoose timestamps]
updatedAt	Date	---	REQ	Last document modification audit timestamp [def: Mongoose timestamps]

DevOpsArtifact Collection

Collection: devopsartifacts • Domain: DevOps & Infrastructure

Mongoose Options: timestamps: true, versionKey: false

Description:

Persistent Dockerfiles, CI/CD pipeline workflows, and Kubernetes manifests generated from repository intelligence.

Database Indexes & Constraints:

- _id (PK)
- { projectId: 1, artifactType: 1, version: -1 } (latest retrieval)
- { projectId: 1, userId: 1 } (tenant isolation index)
- projectId (asc)
- userId (asc)
- artifactType (asc)
- generationStatus (asc)

Attribute Definitions:

KEY	NULL	FIELD NAME	TYPE	DESCRIPTION & DEFAULTS
_id	ObjectId	PK	REQ	Auto-generated unique BSON identifier [def: ObjectId()]
projectId	ObjectId	FK	REQ	Target repository [Physical Reference to Project._id] [def: Indexed]
userId	String	FK	REQ	Clerk User ID [Logical Reference to User.clerkId] [def: Indexed]
artifactType	String	IDX	REQ	Artifact type enum: ['dockerfile', 'cicd', 'kubernetes'] [def: Indexed]
filePath	String	---	REQ	Suggested target file path (e.g. 'Dockerfile') [def: Trimmed]
content	String	---	REQ	Full manifest content (Dockerfile syntax / YAML) [def: Required]
validation	Subdocument	EMB	REQ	Validation diagnostic subdocument [1:1 Validation] [def: {...}]
generationStatus	String	IDX	REQ	Status enum: ['pending','generating','completed','failed'] [def: 'completed']
sourceAnalysisVersion	Mixed	---	REQ	Analysis version used as generation baseline [def: 1]
version	Number	---	REQ	Artifact revision iteration counter [def: 1]
superseded	Boolean	---	REQ	Flag indicating whether newer version exists [def: false]
metadata	Subdocument	---	OPT	Configuration: { runtime, framework, port, provider } [def: {}]
createdAt	Date	---	REQ	Document creation audit timestamp [def: Mongoose timestamps]
updatedAt	Date	---	REQ	Last document modification audit timestamp [def: Mongoose timestamps]

Embedded Subdocuments Specification

Embedded subdocuments represent denormalized child schemas nested directly inside MongoDB documents. ArchMind AI utilizes embedded subdocuments where data has strong lifecycle coupling and bounded growth.

TreeNode

1 : 0..N (Embedded)

Parent Path: Project.tree

Sanitized repository file tree hierarchy with noise-filtered paths, sizes, and Git SHAs.

Fields: path (String, req), type (String: 'file'|'dir', req), size (Number, def: 0), sha (String, opt)

TopExtension

1 : 0..N (Embedded)

Parent Path: Project.stats.topExtensions

Summary file extension distribution tally generated during initial RIE scanner indexing.

Fields: extension (String, req), count (Number, req)

Message

1 : 0..N (Embedded)

Parent Path: Conversation.messages

Chronological chat message document containing role, markdown text, citations, and token metadata.

Fields: messageId (String, req), role (String: 'user'|'assistant', req), content (String, req), citations ([Citation]), metadata (Mixed)

Citation

1 : 0..N (Embedded)

Parent Path: Conversation.messages.citations

Ground-truth source code citation linking LLM assertions to exact file paths and line ranges.

Fields: filePath (String, req), startLine (Number), endLine (Number), sourceType (enum), snippet (String)

SourceLocation

1 : 0..N (Embedded)

Parent Path: SourceFile.sourceLocations

Symbol location index mapping functions, classes, routes, and models to exact source line spans.

Fields: name (String, req), kind (enum: function|class|route|model|variable|entrypoint, req), lineStart (Number, req), lineEnd (Number, req)

Validation & ValidationCheck

1 : 1 & 1 : 0..N (Embedded)

Parent Path: DevOpsArtifact.validation

Multi-tier syntax, structural, and schema validation diagnostics for generated Dockerfiles and YAML.

Fields: valid (Boolean, req), errors ([String]), warnings ([String]), checks ([ValidationCheck]), validationLevel (enum)